

1D CNN — Architecture Variant Comparison

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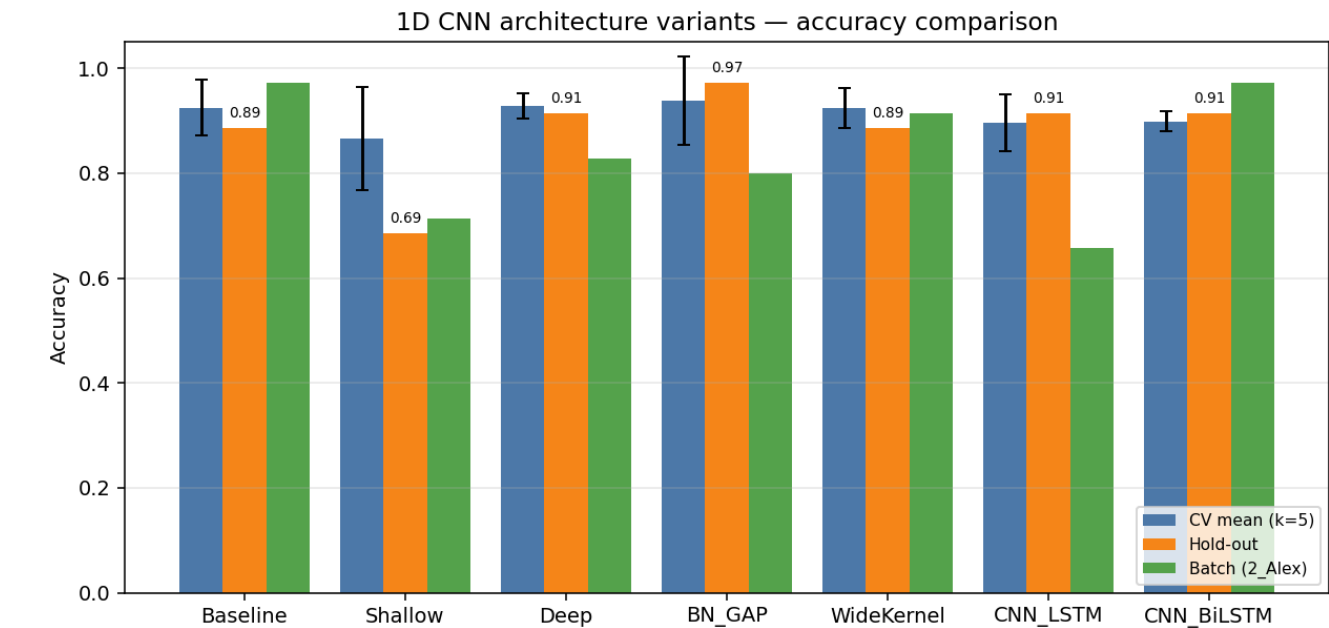
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Total trials	341 (train pool 306, hold-out 35)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

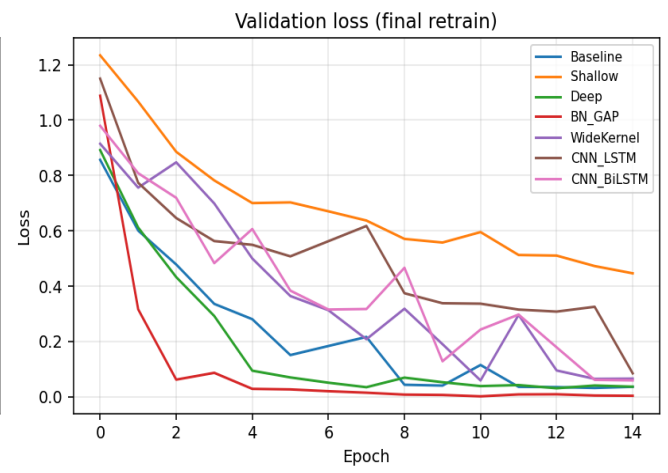
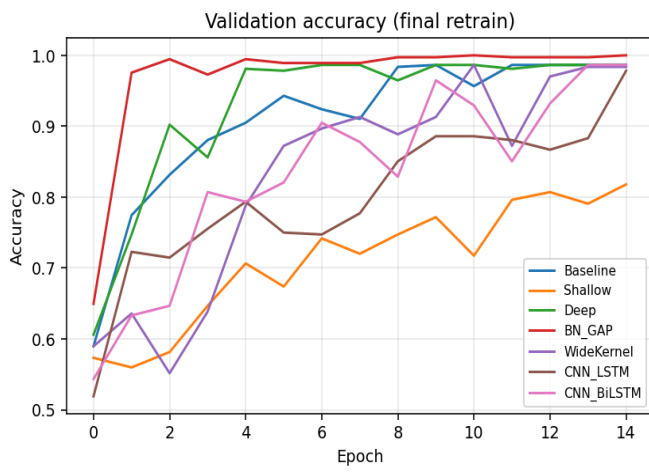
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9248	0.0525	0.8857	0.2537	0.9714	34/35
Shallow	72,487	0.8658	0.0987	0.6857	0.6434	0.7143	25/35
Deep	204,007	0.9280	0.0247	0.9143	0.0956	0.8286	29/35
BN_GAP	57,319	0.9378	0.0845	0.9714	0.0523	0.8000	28/35
WideKernel	131,687	0.9247	0.0383	0.8857	0.3096	0.9143	32/35
CNN_LSTM	68,455	0.8957	0.0544	0.9143	0.2952	0.6571	23/35
CNN_BiLSTM	105,575	0.8987	0.0192	0.9143	0.2305	0.9714	34/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9248 ± 0.0525
Hold-out test acc / loss	0.8857 · 0.2537
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9516	0.9355
2	1470	61	0.9180	0.9344	0.9180
3	1470	61	1.0000	1.0000	1.0000
4	1470	61	0.9344	0.9344	0.9344
5	1470	61	0.8361	0.9836	0.8361

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.667	0.400	0.500	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.899	0.886	0.880	35
weighted avg	0.899	0.886	0.880	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8658 ± 0.0987
Hold-out test acc / loss	0.6857 · 0.6434
Batch-test acc	0.7143 (25/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9516	0.9355
2	1470	61	0.7377	0.8033	0.7377
3	1470	61	0.9672	0.9672	0.9672
4	1470	61	0.7541	0.7541	0.7541
5	1470	61	0.9344	0.9672	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.333	1.000	0.500	5
Double_Nothing	0.667	0.400	0.500	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.000	0.000	0.000	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.600	0.750	5
macro avg	0.714	0.686	0.663	35
weighted avg	0.714	0.686	0.663	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	0	5	0.000
Double_cmere	4	5	0.800
Drum_Roll	5	5	1.000
Overall	25	35	0.714

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9280 ± 0.0247
Hold-out test acc / loss	0.9143 · 0.0956
Batch-test acc	0.8286 (29/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9677	0.9839	0.9677
2	1470	61	0.9016	0.9344	0.9016
3	1470	61	0.9344	0.9836	0.9344
4	1470	61	0.9016	0.9672	0.9016
5	1470	61	0.9344	0.9836	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.400	0.571	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.946	0.914	0.906	35
weighted avg	0.946	0.914	0.906	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	29	35	0.829

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9378 ± 0.0845
Hold-out test acc / loss	0.9714 · 0.0523
Batch-test acc	0.8000 (28/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9839	0.9839	0.9839
2	1470	61	0.7705	0.9344	0.7705
3	1470	61	1.0000	1.0000	1.0000
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9672	0.9836	0.9672

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	0	5	0.000
Double_Raise	3	5	0.600
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	28	35	0.800

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9247 ± 0.0383
Hold-out test acc / loss	0.8857 · 0.3096
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9677	0.9516
2	1470	61	0.9180	0.9344	0.9180
3	1470	61	0.9508	0.9672	0.9508
4	1470	61	0.8525	0.9016	0.8525
5	1470	61	0.9508	0.9508	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.667	0.400	0.500	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.899	0.886	0.880	35
weighted avg	0.899	0.886	0.880	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	2	5	0.400
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8957 ± 0.0544
Hold-out test acc / loss	0.9143 · 0.2952
Batch-test acc	0.6571 (23/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8065	0.8226	0.8065
2	1470	61	0.9180	0.9180	0.9180
3	1470	61	0.9672	0.9672	0.9672
4	1470	61	0.9180	0.9180	0.9180
5	1470	61	0.8689	0.8689	0.8689

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.400	0.571	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	0.833	1.000	0.909	5
macro avg	0.935	0.914	0.902	35
weighted avg	0.935	0.914	0.902	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	3	5	0.600
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	3	5	0.600
Double_Raise	4	5	0.800
Double_Wiggle	1	5	0.200
Double_cmere	4	5	0.800
Drum_Roll	3	5	0.600
Overall	23	35	0.657

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8987 ± 0.0192
Hold-out test acc / loss	0.9143 · 0.2305
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9032	0.9355	0.9032
2	1470	61	0.8689	0.9180	0.8689
3	1470	61	0.9180	0.9180	0.9180
4	1470	61	0.8852	0.9180	0.8852
5	1470	61	0.9180	0.9508	0.9180

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.400	0.571	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.946	0.914	0.906	35
weighted avg	0.946	0.914	0.906	35

Batch test per class (2_Alex)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971